

AYUSHI BHATI

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PROFESSIONAL SUMMARY

Final-year Electronics and Computer Engineering student with hands-on backend and application security exposure in a high-scale fintech environment, including payments and chargeback processing systems, secure coding practices, and vulnerability-aware development. Strong foundation in distributed systems, operating system internals, multi-threaded concurrency, and secure software development life cycle (SSDLC) principles. Skilled in C++, Java, Python, and JavaScript/TypeScript, with practical experience building multi-threaded servers, socket-based networking, CI/CD pipelines, and Docker and Kubernetes based deployments. Microsoft Certified in Azure Fundamentals (AZ-900) and Security, Compliance, and Identity Fundamentals (SC-900), pursuing CompTIA Security Plus, with a track record of improving processing efficiency, system reliability, and secure operational continuity in production-grade systems.

CORE COMPETENCIES

Application Security | Secure Software Development Life Cycle (SSDLC) | Vulnerability Triage and Remediation | Distributed Systems | Backend Development | Payments and Chargeback Processing | Operating System Internals | Multi-Threaded Concurrency | TCP/IP Socket Programming | Secure Coding Practices | API Optimization | System Design | Docker, Kubernetes and Containerization | CI/CD Pipelines | Microservices Architecture | Security Automation and Tooling | Azure Cloud Fundamentals | Cross-Functional Collaboration

TECHNICAL SKILLS

Programming Languages: C++, Java, Python, C, JavaScript, TypeScript, SQL, Go (working knowledge)

Application Security: Secure coding practices, SSDLC, vulnerability triage and remediation, threat-alert logging pipelines, malformed-input and packet handling, AES-GCM encryption/decryption, OAuth 2.0 / JWT access control, role-based access control (RBAC), security automation tooling, reporting dashboards

Distributed and Backend Systems: Multi-threaded concurrency, Inter-Process Communication (IPC), OOP, raw socket programming, TCP/IP, Spring Boot, FastAPI, Node.js, Express.js, REST APIs, Microservices

Cloud and DevOps: Microsoft Azure (AZ-900 certified), Docker, Kubernetes, CI/CD pipelines, Cron/APScheduler, Linux/Unix systems, Bash/Zsh scripting

Networking: ARP, BGP, OSPF, TCP/IP sockets, raw socket programming

Security and Compliance: AES-GCM encryption, malformed-input handling, threat-alert logging, OAuth 2.0 / JWT, security governance, compliance frameworks, identity management (SC-900 certified)

Data Science and Analytics: XGBoost, scikit-learn, SHAP, feature engineering, Pandas, NumPy, model validation

Frontend and Databases: React.js (Hooks), HTML5, CSS3, Tailwind CSS, MongoDB Atlas, PostgreSQL

PROFESSIONAL EXPERIENCE

Backend Development Intern, Engineering Team, Paytm Payments Services Limited **June 2026 - Present**
Noida, India

- Contribute to backend architecture and distributed database infrastructure supporting enterprise-grade fintech workflows that process millions of transactions daily across India.
- Work on payments and chargeback processing systems, applying secure coding practices, API optimization, and high-volume data management to maintain zero-fault operational continuity across production systems.
- Build and maintain CI/CD pipelines and containerize services with Docker to streamline deployment and improve release reliability, in line with secure software development life cycle (SSDLC) practices.
- Analyze large-scale distributed system design patterns to identify and reduce transaction bottlenecks, improving processing efficiency and platform security and reliability.

Frontend Developer Intern, Kredmint Technologies Private Limited **June 2025 - July 2025**
Noida, India

- Engineered responsive UI interfaces for high-traffic financial platforms, ensuring seamless end-user workflows and zero cross-device layout exceptions.
- Integrated high-frequency REST APIs with dynamic state management to enable real-time, accurate data reconciliation and rendering.
- Participated in Agile sprints, structured code reviews, and technical documentation practices to maintain system reliability and secure coding standards.

PROJECTS

Multi-Threaded HTTP Web Server from Scratch

- Programmed a standalone HTTP/1.1 web server, managing networking natively at the OS layer via Linux/macOS sockets to bypass modern framework abstractions.
- Engineered a worker thread pool model to manage socket lifecycles, eliminate memory leaks, and safely handle overlapping client requests concurrently, applying secure resource-handling principles directly relevant to backend platform reliability.

SmartProperty AI - Real Estate Price Prediction and Data Pipeline

- Designed an end-to-end automated machine learning pipeline analyzing 265,500+ property records across 14 cities from multiple fragmented datasets.
- Engineered 72 unique data features, including target encoding and geospatial proximity calculations, achieving R-squared of 0.89 test accuracy and R-squared of 0.87 (+/- 0.02) across 5-fold cross-validation.
- Automated continual data ingestion and cleansing routines, deploying a low-latency backend paired with a real-time monitoring dashboard.

Hotel Booking and Management Platform - Enterprise System

- Architected a dual-interface web application on the MERN stack with secure role-based access control (JWT and OAuth 2.0) to prevent unauthorized privilege escalation.
- Developed full CRUD dashboards with real-time state synchronization, ensuring data consistency and secure handling of client-backend interactions.

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900) - core cloud concepts: compute, storage, networking, pricing, cloud architecture
- Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900) - identity management, compliance frameworks, security governance
- Cisco Networking Academy: Networking Basics - Course Certificate plus 5 Module Achievement Badges
- CompTIA Security+ (Infrastructure and Threat Analysis) - In Progress

EDUCATION

B.Tech in Electronics and Computer Engineering, JC Bose University of Science and Technology, YMCA2023 - 2027

Faridabad, India | CGPA: 7.7

Class XII (CBSE), J.N. International School **2023**

New Delhi, India | 85.2%

Class X (CBSE), Modern Vidya Niketan Sr. Sec. School **2021**

Faridabad, India | 96.8%

SOFT SKILLS AND LANGUAGES

- Cross-Functional Communication: translating complex technical concepts into clear, actionable insights for engineering and security stakeholders.
- Agile Teamwork and Collaboration: collaborating seamlessly within multidisciplinary production teams across multiple time zones.
- Problem-Solving and Attention to Detail: auditing complex systems, triaging vulnerabilities, and troubleshooting high-pressure exceptions.
- Languages: English (Professional Proficiency), Hindi (Native / Bilingual Proficiency)